MIL-E-36890
2 November 1972

MILITARY SPECIFICATION

ESOPHAGOSCOPE, JESBERG, FIBER OPTIC, OVAL

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers fiber optic esophagoscopes, of the Jesberg design, suitable for the examination of the esophagus.

1.2 Classification. Esophagoscopes shall be of the following sizes as indicated below (see 6.1):

Size	Length
1	29 cm
2	50 cm

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposals, form a part of the specification to the extent specified herein:

SPECIFICATIONS

Federal

GG-I-526	Instruments, Dental and Surgical; General Specifications.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

FSC 6515

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

Military

MIL-P-116	Preservation, Methods of
MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof or Waterproof, Flexible.

STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

DRAWING

Defense Personnel Support Center (DPSC)
24307 Esophagoscope, Jesberg, Fiber Optic, Oval

(Copies of specifications, standards, and drawing required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Material. The esophagoscope proper, including the main tube, tip, handle and thumb-nut shall be fabricated from class 8, type 752 material in accordance with Federal Specification GG-I-526, except that the components shall not be plated. The light carrier and light carrier channel shall be fabricated from suitable unplated, nickel-copper alloy (Monel). The bayonet connection of the light carrier shall be fabricated from class 3, type 302, 303, or 304 material in accordance with GG-I-526. The light carrier shall be filled with suitable glass fibers to disperse illumination. Class 3 material shall be subjected to the boil test as specified in GG-I-526. All other material shall be subjected to the boil test and the salt-spray test as specified in GG-I-526.

3.2 Style, design, and dimensions. Shall be a fiber optic esophagoscope of the Jesberg design, suitable for the examination of the interior of the esophagus when assembled to Federal Stock Numbers 6515-104-8434 CABLE, FIBER OPTIC, ENDOSCOPIC INSTRUMENT, 72 inches Long, and 6515-104-8520 LIGHT, ENDOSCOPSIC INSTRUMENT, FIBER OPTIC, 115 Volt, 60 Hz,AC . The instrument proper shall have an oval lumen, thickened at the distal

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:31Z
Check the source to verify that this is the currentversion before use.

end to form a protective edge and shaped to a smooth distal working end. Unit shall be designed to be entered easily and without trauma over the posterior esophageal wall. Instrument shall be furnished with a completely sealed, bayonet-type, light carrier. The light carrier shall be filled with glass fibers to disperse necessary illumination into the esophagus when used with the accessory Federal Stock Numbers specified above. The light carrier shall be designed with a thumb-nut device to securely lock the carrier to the light carrier channel, after the light carrier is placed into the light carrier channel. The male thread that accepts the thumb screw shall be designed so that the thumb screw can be completely removed. The light carrier channel shall be completely sealed at the outer surface of the distal end. When the light carrier is assembled to the carrier channel, the lumen of the instrument shall not be obstructed. The handle and light carrier channel shall be permanently secured to the main tube. The nominal lumen of the esophagoscopes shall be 10 millimeters (mm) in the horizontal axis, and 14 mm in the vertical axis. The complete instrument shall be in accordance with DPSC Drawing 24307 for the appropriate Size and Federal Stock Number.

3.3 Finish. The complete instrument shall have a smooth finish throughout. Burrs, cracks, holes, pits, corrosion, and sharp edges shall be prohibited. The enlarged distal edge of the instrument shall be smoothly contoured to the main tube.

3.4 Instrument operation. When tested as specified in 4.4.1, the light carrier shall be capable of adequate insertion into the light carrier channel and be secured to the handle by the thumb-screw. Further, illumination shall be readily observed at the distal end of the light carrier.

3.5 Identification marking. The instrument shall be permanently marked with the name (or registered trademark) of the manufacturer, size, and country of origin. Further, the light carrier shall bear permanent length markings (29 for Size 1, and 50 for Size 2) and the name or registered trademark of the manufacturer.

3.6 Container. Each instrument shall be furnished in a suitable container, cushioned to prevent damage to instrument during transit.

3.7 Instructions. A detailed instruction sheet for proper use and maintenance of the esophagoscopes shall be furnished with each instrument.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

3.8 Workmanship. Workmanship shall be first class throughout. The instruments shall be free from defects which detract from their appearance or impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which such records relate.

4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing determination by technical means of physical and chemical properties) of the item.

4.1.3 Certificates of quality. Certificates of quality, supplied by the manufacturer of the nickel-silver and nickel-copper alloys, may be furnished in lieu of actual performance of such testing by the contractor, provided lot identity has been maintained and can be demonstrated to the Government. The certificate shall include the name of the contractor, the contract number, the name of the manufacturer or supplier, the FSN, the item identification, the name of the component/material, the lot number, the lot size, the sample size, the date of testing, the test method, individual test results, and the specification requirements.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105, and as indicated in Table I. The unit of product for examination purposes shall be one complete esophagoscope.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

Table I. Sampling Criteria for Examination

Type defect	Inspection level	AQL (percent defective)
Critical	100 percent	No. defects permitted
Major	II	1.0
Minor	II	2.5

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105, and as indicated in Table II. The unit of product for testing purposes shall be one complete esophagoscope. Where feasible, the same sample unit shall be used for the determination of two or more test characteristics.

Table II. Sampling Criteria for Tests

Component	Characteristic	Requirement	Test procedure	Inspection level	AQL
Esophagoscope	Composition analysis	3.1 and GG-I-526	GG-I-526	*	-
	Boil test	3.1 and GG-I-526	GG-I-526	S-1	1.0
	Salt-spray test	3.1 and GG-I-526	GG-I-526	S-1	1.0
	Dimensions	3.2	____	S-1	1.5
	Operation test	3.4	4.4.1	100%	**

* Sufficient quantity of each lot of each metal used in the construction of the esophagoscopes shall be selected for chemical analysis.
** No defects permitted.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

4.3 Examination. The esophagoscopes shall be examined to determine compliance with all the requirements contained in this specification. Examination shall be conducted as specified in Tables I and III. Examination, however, shall not be restricted to defects listed in Table III. Any instruments in the sample containing one or more defects shall be rejected, and if the number of defective instruments in any sample exceeds the acceptance number for that sample, the lot represented by that sample shall be rejected.

Table III. Classification of Defects

Categories	Defects
Critical	
01	Outer surface of instrument proper not free from sharp edges and burrs.
02	Handle and light carrier channel not permanently secured to main tube.
Major	
101	Esophagoscope not of the specified design and size.
102	Light carrier not furnished with unit.
103	Light carrier not of bayonet design.
104	Glass fibers not completely encased and sealed in light carrier housing.
105	Lumen of instrument not oval in cross section.
106	Finish of outer surface and lumen not free from cracks, holes, pits, burrs, sharp edges and corrosion.
107	Lumen obstructed when light carrier is assembled to light carrier channel.
108	Components plated.
109	Light carrier channel not completely sealed at outer surface of distal end.
110	Distal end not smoothly blended to outside surfaces of main tube.
111	Thumb-screw can be removed from male thread by hand.
112	Size markings not present.
Minor	
201	Manufacturer's name and country of origin not present on instrument.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Operation test. This test shall be performed on a 100 percent basis. Insert the light carrier into the light carrier channel. Secure the light carrier to instrument handle by means of the thumb-screw. Failure to adequately insert and secure the light carrier to instrument shall be cause for immediate rejection. Assemble fiber optic cable, with bayonet connection (FSN 6515-104-8434) to bayonet portion of light carrier. Assemble fiber optic cable to endoscopic light source (FSN 6515-104-8520). With the light source properly connected to 115 Volt, 60 H_z, AC power source, observe the distal end of light carrier for illumination. Illumination shall not flicker during the test.

5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One esophagoscope, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in Table IV shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table IV. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	12 units	24 units

5.1.2.1 Packing variation permitted. If the required number of units in the entire shipment is less than the number of units specified to be overpacked in an exterior container, such units may be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Cleaning and drying. Each unit shall be thoroughly cleaned and dried in accordance with the applicable method as outlined in MIL-P-116.

5.1.3.2 Preservative. Not required.

Source: https://assist.dla.mil -- Downloaded: 202640-21T09:31Z
Check the source to verify that this is the current version before use.

5.1.3.3 Intimate wrap. Each unit shall be wrapped in neutral tissue or other suitable anticorrosive material.

5.1.3.4 Unit package. Each unit shall be packaged in accordance with MIL-P-116, method IC-2.

5.1.3.5 Intermediate package. Intermediate package shall be a box of appropriate size constructed in accordance with PPP-B-676, except commercial colors will be acceptable, or PPP-B-636, type CF, class domestic. Closure shall be adequate to prevent accidental opening under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Fiberboard boxes shall conform to special requirements specified in PPP-B-636.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:31Z
Check the source to verify that this is the current version before use.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall be designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, class domestic. Closure of wood boxes shall be as specified in the appendix of the applicable box specification. Closure of fiberboard boxes shall conform to method II of PPP-B-636. In addition, fiberboard boxes shall conform to the special requirements specified in PPP-B-636.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Purchasers should exercise any desired options offered herein, and the procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Size required (see 1.2).
(c) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 This specification is intended to cover the following items appearing in the Federal Supply Catalog:

Federal Stock No.	Item Identification
6515-104-8679	ESOPHAGOSCOPE, Jesberg, Fiber Optic, Oval, 10 mm by 14 mm by 29 cm.

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.

Federal Stock No.	Item Identification
6515-104-8680	ESOPHAGOSCOPE, Jesberg, Fiber Optic, Oval, 10 mm by 14 mm by 50 cm.

It does not include all types or sizes as indicated by the title nor all types which are commercially available.

Custodians:

Army - MD
Navy - MS
Air Force - 03

Review activities:

Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Project No. 6515-1665

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:31Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-E-36890

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A
☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

Source: https://assist.dla.mil -- Downloaded: 202610-21T09:31Z
Check the source to verify that this is the current version before use.

FOLD

DEFENSE PERSONNEL SUPPORT CENTER
2800 South 20th Street
Philadelphia, Pa. 19101

Defense Supply Agency

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

POSTAGE AND FEES PAID
DEFENSE SUPPLY AGENCY
DoD - 304

U.S.MAIL

HEADQUARTERS, DEFENSE PERSONNEL SUPPORT CENTER
ATTN: DIRECTORATE OF MEDICAL MATERIEL, CODE ATT
2800 SOUTH 20TH STREET
PHILADELPHIA, PA. 19101

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-21T09:31Z
Check the source to verify that this is the current version before use.